**Institute of Engineering, Thapathali Campus**
**Department of Electronics and Computer Engineering**
**Computer Network Lab Manual**

## LAB MANUAL 8 : Static and Default Routing in a Multi-Router Network

### 1. Lab Objectives

- Design and build a multi-router topology from scratch in Cisco Packet Tracer.
- Configure IPv4 addressing on LAN interfaces, WAN point-to-point links, and loopback interfaces using the Cisco IOS CLI.
- Implement manual **Static Routing** to establish bidirectional data flow between isolated networks.
- Configure a **Quad-Zero Default Route** (0.0.0.0/0) to simulate an exit gateway to the Internet.
- Diagnose and resolve routing failures using core verification tools (ping, traceroute, show ip route).

### 2. Topology Design & Addressing Table

Place three routers, two switches, and two end-devices into the logical workspace. Use **Cisco 4331 Routers** or **2911 Routers**. If your routers do not have enough built-in serial or Ethernet ports, turn off the router power switch in the physical tab, drag the **NIM-2T** modules into an open slot, and flip the power switch back on.

**Topology Layout**

- **PC-HQ** → **Switch-HQ** → **Router-HQ** (Gig0/0/0)
- **Router-HQ** (Se0/1/0) → Directly connected to → **Router-BR** (Se0/1/0)
- **Router-BR** (Gig0/0/0) → **Switch-BR** → **PC-BR**
- **Router-BR** (Se0/1/1) → Directly connected to → **Router-ISP** (Se0/1/0)
- **Router-ISP** uses a virtual Loopback interface to simulate public web servers (e.g., 8.8.8.8).

**Addressing Table**

| Device | Interface | IP Address | Subnet Mask | Default Gateway |
|---|---|---|---|---|
| **Router-HQ** | Gig0/0/0 (LAN) | 192.168.10.1 | 255.255.255.0 | N/A |
| | Se0/1/0 (WAN) | 10.1.1.1 | 255.255.255.252 | N/A |
| **Router-BR** | Gig0/0/0 (LAN) | 192.168.20.1 | 255.255.255.0 | N/A |
| | Se0/1/0 (WAN) | 10.1.1.2 | 255.255.255.252 | N/A |
| | Se0/1/1 (ISP WAN) | 203.0.113.1 | 255.255.255.252 | N/A |
| **Router-ISP** | Se0/1/0 (WAN) | 203.0.113.2 | 255.255.255.252 | N/A |
| | Loopback0 (Server) | 8.8.8.8 | 255.255.255.255 | N/A |
| **PC-HQ** | NIC | 192.168.10.10 | 255.255.255.0 | 192.168.10.1 |
| **PC-BR** | NIC | 192.168.20.10 | 255.255.255.0 | 192.168.20.1 |

## 3. Step-by-Step Configuration Guide

### A.Topology Setup and Host Initialization

Assemble the hardware elements in the workspace using appropriate cabling (Copper Straight-Through for device-to-switch; Serial Cable for router-to-router).

Open the desktop configuration tab on **PC-HQ** and **PC-BR** and assign their static IPs, subnet masks, and default gateways exactly as defined in the addressing table. Do not leave the gateway blank, or packets will fail to leave the local broadcast domain.

### B.Basic Interface and Link Configuration

Access the CLI of all three routers. Change their default hostnames to prevent admin confusion, then configure and activate the physical interfaces.

**! Configuration on Router-HQ**
Router> enable
Router# configure terminal
Router(config)# hostname Router-HQ
Router-HQ(config)# interface gigabitEthernet 0/0/0
Router-HQ(config-if)# ip address 192.168.10.1 255.255.255.0
Router-HQ(config-if)# no shutdown
Router-HQ(config-if)# exit

Router-HQ(config)# interface serial 0/1/0
Router-HQ(config-if)# ip address 10.1.1.1 255.255.255.252
Router-HQ(config-if)# no shutdown
Router-HQ(config-if)# exit

**! Configuration on Router-BR**
Router> enable
Router# configure terminal
Router(config)# hostname Router-BR
Router-BR(config)# interface gigabitEthernet 0/0/0
Router-BR(config-if)# ip address 192.168.20.1 255.255.255.0
Router-BR(config-if)# no shutdown
Router-BR(config-if)# exit

Router-BR(config)# interface serial 0/1/0
Router-BR(config-if)# ip address 10.1.1.2 255.255.255.252
Router-BR(config-if)# no shutdown
Router-BR(config-if)# exit

Router-BR(config)# interface serial 0/1/1
Router-BR(config-if)# ip address 203.0.113.1 255.255.255.252
Router-BR(config-if)# no shutdown
Router-BR(config-if)# exit

**! Configuration on Router-ISP**
Router> enable
Router# configure terminal
Router(config)# hostname Router-ISP
Router-ISP(config)# interface serial 0/1/0
Router-ISP(config-if)# ip address 203.0.113.2 255.255.255.252
Router-ISP(config-if)# no shutdown
Router-ISP(config-if)# exit

**! Create a virtual interface to simulate external server infrastructure**
Router-ISP(config)# interface loopback 0
Router-ISP(config-if)# ip address 8.8.8.8 255.255.255.255
Router-ISP(config-if)# exit

### C.Pre-Routing Verification Analysis

Before implementing routing, examine the current isolation state of the network layers.

Execute the following commands on **Router-HQ**:

Router-HQ# show ip interface brief
Router-HQ# show ip route

Verify that only **directly connected networks (C)** and **local host paths (L)** appear in the routing matrix.

Try to ping **PC-BR** (192.168.20.10) from **PC-HQ** (192.168.10.10). The ping must fail. Note down the explicit error message displayed in the command line terminal.

### D.Bidirectional Static Route Injection

Inject static routes so the Headquarter LAN and Branch LAN can communicate. Use the **Next-Hop IP address** method to explicitly declare the entry gateway of the neighboring autonomous hop.

**! On Router-HQ: Direct traffic bound for the Branch LAN to Router-BR's incoming WAN IP**

Router-HQ(config)# ip route 192.168.20.0 255.255.255.0 10.1.1.2

**! On Router-BR: Direct traffic bound for the Headquarter LAN to Router-HQ's incoming WAN IP**

Router-BR(config)# ip route 192.168.10.0 255.255.255.0 10.1.1.1

Re-test connectivity by running a ping from **PC-HQ** to **PC-BR**. The transmission should now succeed. If the initial packet drops, analyze the ARP registration process that occurred over the link.

**E.Default Route Integration (Simulating Internet Traffic)**

Right now, neither HQ nor Branch can communicate with the simulated internet server (8.8.8.8). Instead of writing unique static routes for every new IP on the internet, implement a **Default Route** strategy.

**! On Router-HQ: Point a default route to Router-BR**
Router-HQ(config)# ip route 0.0.0.0 0.0.0.0 10.1.1.2

**! On Router-BR: Point a default route out to the ISP edge**
Router-BR(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.2

**! On Router-ISP: The ISP needs a return route to know how to reach internal networks**
**! We will create a summary static route to cover both internal subnets (192.168.10.0 and 192.168.20.0)**

Router-ISP(config)# ip route 192.168.0.0 255.255.0.0 203.0.113.1

**F.End-to-End Validation and Trace Diagnostics**
Confirm complete structural convergence across the entire multi-tier system.
From **PC-HQ**, open the command prompt interface and execute:

C:\> ping 8.8.8.8
C:\> tracert 8.8.8.8

Ensure that the traceroute mapping successfully logs three distinct layer hops before reading the destination address.Finally, secure the dynamic running memory maps to the permanent NVRAM across all control modules:

Router# copy running-config startup-config

**4. Guided Troubleshooting Scenarios**

- **Scenario A: Asymmetric Link Rupture**
  - *Action:* On Router-BR, remove the static route pointing back to the HQ LAN (no ip route 192.168.10.0 255.255.255.0 10.1.1.1).
  - *Diagnostic Task:* Run a ping from PC-HQ to PC-BR. Switch to Packet Tracer's **Simulation Mode** and trace the envelope icon. They will see the packet successfully arrive at PC-BR, but drop on the return leg at Router-BR. This visually reinforces that routing paths must always be bidirectional.
- **Scenario B: The Subnet Mask Trap**
  - *Action:* Intentionally misconfigure a static route path mask to point to a /30 instead of a /24 (e.g., ip route 192.168.20.0 255.255.255.252 10.1.1.2).
  - *Diagnostic Task:* Check why PC-HQ can ping 192.168.20.1 but completely fails to reach 192.168.20.10. This shows them how specific route prefix matching works.

**5. Post-Lab Assessment & Report Deliverables**

Students must include answers to the following engineering design questions in their final lab submission:

1. **Administrative Distance Evaluation:** Look at the output of show ip route. What does the notation [1/0] mean next to the static routes? If you configure a dynamic routing protocol like OSPF on these exact links later, which path will the router prefer, and why?
2. **Next-Hop vs. Exit-Interface:** If you configure a static route using an exit interface (e.g., ip route 192.168.20.0 255.255.255.0 Gig0/0/1) instead of a next-hop IP on a multi-access network like Ethernet, what hazardous performance side effects occur regarding ARP storage sizing?
3. **Route Summarization Logic:** In Step 5, the ISP router used a summary route of 192.168.0.0 255.255.0.0. Prove mathematically how this single route entry matches both the 192.168.10.0/24 and 192.168.20.0/24 subnets simultaneously.